

Bidding on multiple advertising campaigns

Online Learning Applications course project - A.Y. 2025/2026

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- 2 Requirement 2: stochastic multiple-campaigns environment
- 3 Requirement 3: highly non-stationary multiple-campaigns environment
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But first... some definitions

- total starting budget: B
- number of users (or, equivalently, of action rounds): T
- per-round budget: $\rho = \frac{B}{T}$
- set of available bids: $\mathcal{B}$
- bidding probability distribution over bidder actions (bids): $\gamma \in \Delta_{\mathcal{B}}$
- probability distribution of the highest competing bid of a campaign: $\mathcal{D}_m$
- utility: $f(b) = (v - b) \mathbb{1}[b \geq m]$
- cost: $c(b) = b \mathbb{1}[b \geq m]$
- expected utility induced by the highest competing bids probability distribution: $\bar{f}(b) = \mathbb{E}_{m \sim \mathcal{D}_m} [f(b)]$
- expected cost induced by the highest competing bids probability distribution: $\bar{c}(b) = \mathbb{E}_{m \sim \mathcal{D}_m} [c(b)]$
- expected utility induced by the bidding probability distribution: $f(\gamma) = \mathbb{E}_{b \sim \gamma} [f(b)]$
- expected cost induced by the bidding probability distribution: $c(\gamma) = \mathbb{E}_{b \sim \gamma} [c(b)]$
- overall expected utility: $\bar{f}(\gamma) = \mathbb{E}_{b \sim \gamma} [\mathbb{E}_{m \sim \mathcal{D}_m} [(v - b) \mathbb{1}[b \geq m]]] = \mathbb{E}_{b \sim \gamma} [\mathbb{E}_{m \sim \mathcal{D}_m} [f(b)]]$
- overall expected cost: $\bar{c}(\gamma) = \mathbb{E}_{b \sim \gamma} [\mathbb{E}_{m \sim \mathcal{D}_m} [c(b)]] = \mathbb{E}_{b \sim \gamma} [\mathbb{E}_{m \sim \mathcal{D}_m} [b \mathbb{1}[b \geq m]]]$

Requirement 1: stochastic single-campaign environment

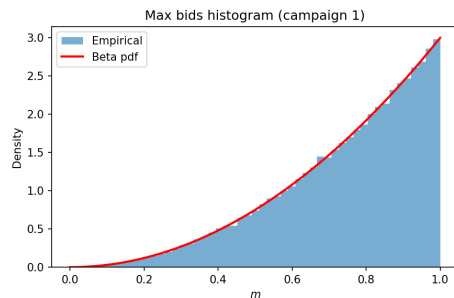
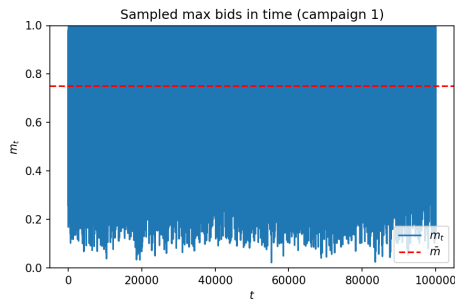
Stochastic single-campaign environment

- One campaign, T sequential first-price auctions
- Stochastic environment = stationary stochastic process, with a constant-across-rounds distribution $\mathcal{D}_m$ of the highest competing bid m (unknown by the bidder)
- Per-round budget $\rho = B/T$, total budget B (possibly $+\infty$ to model no-budget setting)
- Valuation v of the bidder
- Two settings to be studied:
 - **UCB-like** without budget
 - **UCB-like** with budget

Competing-bid distribution: from uniform to Beta

Modeling choice:

- n competitors draw bids $X_i \sim \text{Unif}(0, 1)$
- Maximum $m = \max_i X_i \sim \text{Beta}(n, 1)$
- PDF: $\mathbb{P}[m = b] = nb^{n-1}$
- CDF (win probability): $\mathbb{P}[m \leq b] = b^n$



Clairvoyant

A clairvoyant "in expectation":

- the highest competing bids m_t are stochastic and sampled from a distribution $\mathcal{D}_m$
 $\implies$ we need the best behavior *on average*
- the clairvoyant only needs to satisfy the budget constraint *in expectation*, but is allowed to exceed per-round budget at any round

Assumption

The clairvoyant knows the underlying distribution $\mathcal{D}_m$ of the highest competing bids m_t , but not their actual realizations!

Clairvoyant - example

Consider the following simple setting:

- discretized bids space $\mathcal{B} = \{0.0, 0.4\}$
- ad valuation of the bidder $v = 0.8 \implies$ if you win, you always get a reward
- per-round budget $\rho = 0.2$
- expected highest competing bid $\mathbb{E}_{m \sim \mathcal{D}_m} [m] = 0.1$

Clairvoyant - example

In this setting:

- always bidding 0.0 respects the budget constraint but never wins any auction and thus never gets any value ($f = 0.0$)
- always bidding 0.4 gets some value, but will likely exceed the budget constraint in expectation

⇒ a distribution over bids is needed!

There exists a single optimal bid (by existence of a deterministic policy in a stationary MDP), but we cannot use it because of the discretization of the bids space!

Clairvoyant - definition

Choose an optimal *distribution* $\gamma^*(b)$ over bids that:

maximizes the final expected reward

subject to the budget constraint in expectation

$$\gamma^* = \arg \max_{\gamma \in \Delta_{\mathcal{B}}} \bar{f}(\gamma)$$

$$\text{s.t. } T\bar{c}(\gamma) \leq B$$

$$= \arg \max_{\gamma \in \Delta_{\mathcal{B}}} \bar{f}(\gamma)$$

$$\text{s.t. } \bar{c}(\gamma) \leq \rho$$

$$= \arg \max_{\gamma \in \Delta_{\mathcal{B}}} \mathbb{E}_{b \sim \gamma} [\mathbb{E}_{f \sim \mathcal{D}} [f(b)]]$$

$$\text{s.t. } \mathbb{E}_{b \sim \gamma} [\mathbb{E}_{c \sim \mathcal{D}} [c(b)]] \leq \rho$$

$$= \arg \max_{\gamma \in \Delta_{\mathcal{B}}} \mathbb{E}_{b \sim \gamma} [\mathbb{E}_{m \sim \mathcal{D}_m} [(v - b) \mathbb{1}[b \geq m]]]$$

$$\text{s.t. } \mathbb{E}_{b \sim \gamma} [\mathbb{E}_{m \sim \mathcal{D}_m} [b \mathbb{1}[b \geq m]]] \leq \rho$$

$$= \arg \max_{\gamma \in \Delta_{\mathcal{B}}} \mathbb{E}_{b \sim \gamma} [(v - b) \mathbb{E}_{m \sim \mathcal{D}_m} [\mathbb{1}[b \geq m]]]$$

$$\text{s.t. } \mathbb{E}_{b \sim \gamma} [b \mathbb{E}_{m \sim \mathcal{D}_m} [\mathbb{1}[b \geq m]]] \leq \rho$$

$$= \arg \max_{\gamma \in \Delta_{\mathcal{B}}} \mathbb{E}_{b \sim \gamma} [(v - b) \mathbb{P}_{m \sim \mathcal{D}_m} [b \geq m]]$$

$$\text{s.t. } \mathbb{E}_{b \sim \gamma} [b \mathbb{P}_{m \sim \mathcal{D}_m} [b \geq m]] \leq \rho$$

$$= \arg \max_{\gamma \in \Delta_{\mathcal{B}}} \sum_b \gamma(b) \underbrace{(v - b) \mathbb{P}_{m \sim \mathcal{D}_m} [b \geq m]}_{\bar{f}(b)}$$

$$\text{s.t. } \sum_b \gamma(b) \underbrace{b \mathbb{P}_{m \sim \mathcal{D}_m} [b \geq m]}_{\bar{c}(b)} \leq \rho$$

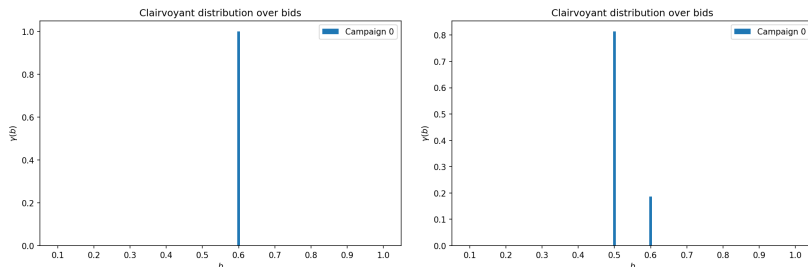
Clairvoyant - Linear Program

By writing the simplex constraints explicitly, we obtain the following Linear Program (LP):

$$\begin{aligned}
 \gamma^* &= \arg \max_{\gamma \in \mathbb{R}^K} && \sum_b \gamma(b) \bar{f}(b) \\
 \text{s.t.} &&& \sum_b \gamma(b) \bar{c}(b) \leq \rho \\
 &&& \sum_b \gamma(b) = 1 \\
 &&& 0 \leq \gamma(b) \leq 1 \quad \forall b
 \end{aligned}$$

Clairvoyant - optimal distribution γ^* over bids

- With budget, γ^* can concentrate on multiple bids, attempting to spend exactly the whole available budget
- **Note:** for $B = +\infty$ the budget constraint disappears $\implies$ we no longer care about the cost and can choose whatever deterministic bid we want, even if we overspend



Left: with no budget, the LP puts all the probability mass on the bid b^* that maximizes $\bar{f}(b) = (v - b) \mathbb{P}(b \geq m)$

Right: with budget, it splits the probability mass on two adjacent bids

UCB-like bidder: algorithm

- Treat each bid $b \in \{0, 0.1, \dots, 1.0\}$ as an arm
- Keep track of empirical means $\hat{f}(b)$, $\hat{c}(b)$ from bandit feedback
- Compute the current confidence radius:

$$r(b) = v \cdot \sqrt{\frac{2 \log t}{N(b)}}$$

- Use it to implement optimism:

$$\hat{f}_{\text{UCB}}(b) = \min\{\hat{f}(b) + r(b), 1\},$$

$$\hat{c}_{\text{LCB}}(b) = \max\{\hat{c}(b) - r(b), 0\}$$

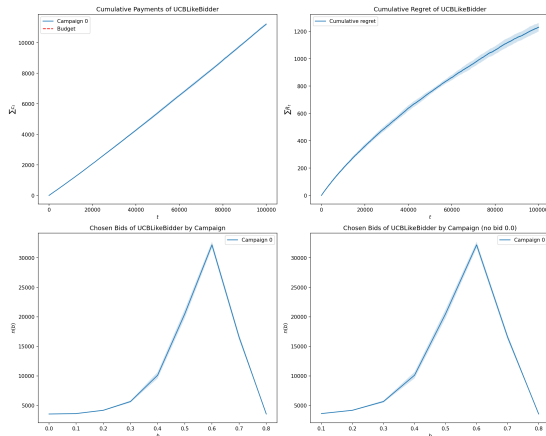
UCB-like bidder: action selection

- **Exploration rounds (first K rounds):** round-robin exploration of all the K bids
- **Action selection:** we can solve the same LP as the clairvoyant, replacing the true $\bar{f}(b)$ and $\hat{c}(b)$ with the optimistic estimates

$$\begin{aligned}
 \gamma^* &= \arg \max_{\gamma \in \mathbb{R}^K} && \sum_b \gamma(b) \hat{f}_{\text{UCB}} \\
 \text{s.t.} &&& \sum_b \gamma(b) \hat{c}_{\text{LCB}} \leq \rho, \\
 &&& \sum_b \gamma(b) = 1 \\
 &&& 0 \leq \gamma(b) \leq 1 \quad \forall b
 \end{aligned}$$

and then sample a bid from the resulting optimal distribution γ^*

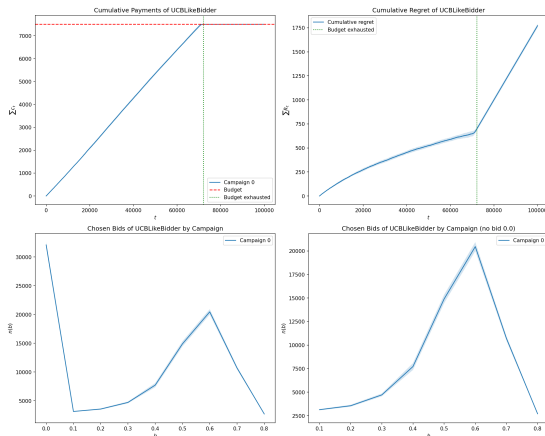
No-budget: empirical behavior



$\rho = \infty$: bid mass concentrates on the single arm maximizing f_{UCB} (matches the no-budget clairvoyant).

- It takes 100000 (!) rounds for the UCB-like bidder to start exhibiting sublinear regret
- This is because the confidence radius decay slowly for an action space of $|\mathcal{B}| = 10$ bids $\implies$ a confidence radius that decays faster should be studied
- Still, the played bids are concentrating around the bid played by the no-budget clairvoyant

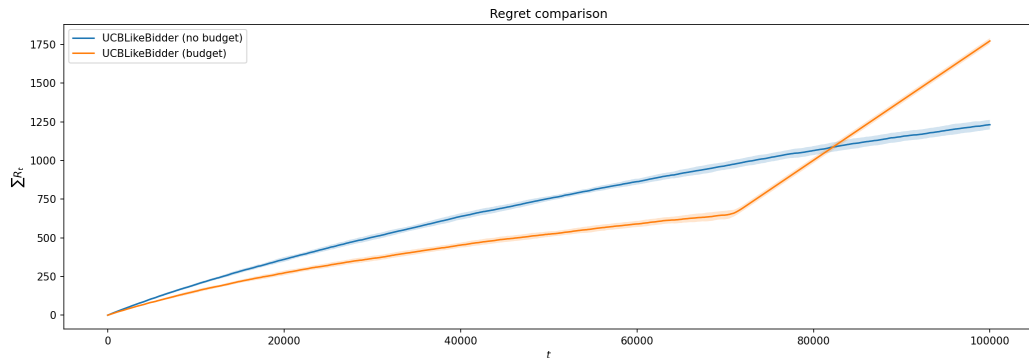
Budget: empirical behavior



- Again, sublinear regret after a lot of rounds
- The peak in the played action is also once again concentrating around the bids played by the no-budget clairvoyant

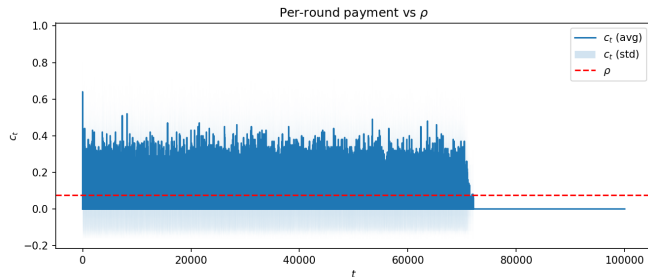
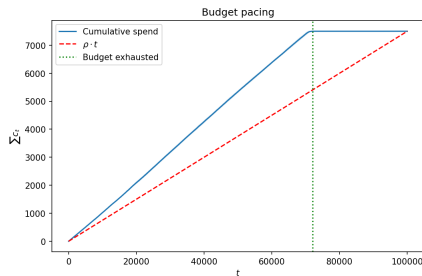
$\rho < \infty$: there are a lot of no-bid actions, because the budget gets depleted. Also, other bids are more spread out

No-budget vs budget: regret comparison



Sublinear regret in both regimes. **With budget**, regret is initially lower because the clairvoyant utility itself is restricted by the budget constraint; however, it eventually skyrockets when the budget is depleted, while the **no-budget** regret keeps growing with a sublinear trend.

Budget pacing



In the budget-constrained setting, we observe the bidder overspending (**left plot**) with respect to the per-round budget ρ : this is likely because of exploration, as it seems to be possible to spend much more than ρ , while it is not possible to spend that much less (**right plot**).

Requirement 2: stochastic multiple-campaigns environment

Stochastic multiple-campaigns environment

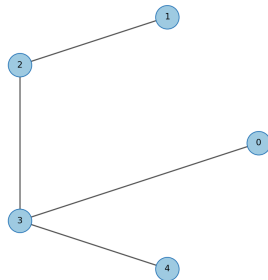
- N campaigns, each with its own ad valuation $v^{(i)}$, and joint highest-bid distribution $\mathcal{D}_m$
- Single shared budget B across *all* campaigns
- Conflict graph $\mathcal{G}$: certain pairs of campaigns are mutually exclusive (cannot win both)
- Action per round: choose N bids, one for each campaign, from a bid space $\mathcal{B}$ of size K
 $\implies K^N$ possible superarms
- Algorithm to be studied: **Combinatorial-UCB** with budget
 - Its output is a distribution γ for the probability of choosing each superarm

Conflict graph - model

A graph in which:

- each node represents a campaign
- an edge between two nodes represents a conflict between incompatible campaigns, meaning that they cannot be bid on during the same round (but both can be bid on in different rounds)

Conflict graph between campaigns



Conflict graph (matrix)

	0	1	2	3	4
0	0	0	0	1	0
1	0	0	1	0	0
2	0	1	0	1	0
3	1	0	1	0	1
4	0	0	0	1	0
Campaign	0	1	2	3	4

Joint distribution of competing bids

- Simplest model: per-campaign highest competing bid $m_t^{(i)} \sim \text{Beta}(n_i, 1)$, sampled independently
 - Corresponds to each competitor bidding independently of others, even across campaigns
 - The joint distribution between all the campaigns can be defined by simply multiplying the product of the independent probabilities corresponding to a set of highest competing bids
- Reality: fully-joint distribution $\mathcal{D}_{\mathbf{m}}$ which models correlations among the highest competing bids $\mathbf{m}$ of the campaigns
 - This is because each competitor may bid across multiple campaigns, introducing correlations (common budget, common objective, common market condition)
 - In this case, the joint distribution cannot be decomposed into single-campaign distribution

Do we really need to model correlations between campaigns?

Structural decomposition of the utilities and costs

Assumption: Without loss of generality, we consider the same ad valuation for all the campaigns $\implies$ the bid space $\mathcal{B}$ is identical (same available bids for all the campaigns)

- At every round, we choose one superarm $\mathbf{b} = (b_1, \dots, b_N) \in \mathcal{B}^N \sim \gamma$ and the environment chooses the highest competing bids $\mathbf{m} = (m_1, \dots, m_N) \sim \mathcal{D}_{\mathbf{m}}$ for each campaign
- We observe a utility $f^{(i)}(b_i, m_i)$ for each campaign; the total utility $f(\mathbf{b}, \mathbf{m})$ received is the sum of the per-campaign utilities:

$$f(\mathbf{b}, \mathbf{m}) = \sum_{i=1}^N f^{(i)}(b_i, m_i)$$

Structural decomposition of the utilities and costs

- The expected utility $\bar{f}(\gamma)$ for any superarm distribution γ and joint highest-competing-bid distribution $\mathcal{D}_{\mathbf{m}}$ is

$$\bar{f}(\gamma) = \mathbb{E}_{\mathbf{b} \sim \gamma} \left[\mathbb{E}_{(m^{(1)}, \dots, m^{(N)}) \sim \mathcal{D}_{\mathbf{m}}} \left[\sum_{i=1}^N f^{(i)}(b_i, m_i) \right] \right]$$

- By linearity of expectation,

$$\bar{f}(\gamma) = \sum_{i=1}^N \mathbb{E}_{\mathbf{b} \sim \gamma} \left[\mathbb{E}_{(m^{(1)}, \dots, m^{(N)}) \sim \mathcal{D}_{\mathbf{m}}} \left[f^{(i)}(b_i, m_i) \right] \right]$$

Structural decomposition of the utilities and costs

- Since each $f^{(i)}$ only depends on the highest competing bid in that same campaign,

$$\bar{f}(\gamma) = \sum_{i=1}^N \mathbb{E}_{\mathbf{b} \sim \gamma} \left[\mathbb{E}_{m^{(i)} \sim \mathcal{D}_{m_i}} \left[f^{(i)}(b_i, m_i) \right] \right] = \sum_{i=1}^N \bar{f}^{(i)}(\gamma)$$

with $\mathcal{D}_{m_i}$ being the **marginalized** distribution for the highest competing bid m_i of campaign i and $\bar{f}^{(i)}(\gamma)$ being its utility for the current action distribution γ over superarms.

$\implies$ correlations among the highest competing bids of all campaigns disappear!

Structural decomposition of the utilities and costs

Takeaway

A joint model having correlations between the highest competing bids $m_1, \dots, m_N$ is not inherently more expressive than a model that independently samples the highest competing bids of the campaigns according to their marginal probabilities according to the joint model, **as far as it concerns our quantities of interest, because they are linear.**

⇒ We can simply model (and also sample) the highest-competing-bid distributions of the campaigns independently!

A few notes:

- The same also holds for the expected cost $\bar{c}(\gamma)$
- This means that we can consider separate values of $f^{(i)}(b_i)$ and $c^{(i)}(b_i)$ (or of their estimates) for each bid b_i of every campaign i , like we did in the single-campaign setting!

Clairvoyant

- A clairvoyant "in expectation":
 - the highest competing bids m_t are stochastic and sampled from a distribution $\mathcal{D}_m$
 $\implies$ we need the best behavior *on average*
 - the clairvoyant only needs to satisfy the budget constraint *in expectation*, but is allowed to exceed per-round budget at any round

Assumption

The clairvoyant knows the underlying distribution $\mathcal{D}_m$ of the highest competing bids m_t , but not their actual realizations!

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Conceptually, the same as the single-campaign clairvoyant!

However, now we have *multiple campaigns...*

What is the optimal strategy?

Assumption

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lowed to

m_t , but

The apparent difficulty: the superarm space

- A superarm is a complete vector of bids:

$$\mathbf{b} = (b_1, \dots, b_N).$$

- With $K = |\mathcal{B}|$ possible bids per campaign, the number of possible superarms is

$$|\mathcal{B}|^N = K^N.$$

- A generic clairvoyant policy would therefore be a distribution

$$\gamma(\mathbf{b}), \quad \mathbf{b} \in \mathcal{B}^N.$$

- This is exponentially large in the number of campaigns and polynomially large in the number of bids.

We have already removed the correlations in the highest competing bids! Can we also avoid optimizing over the full joint distribution?

Why randomization over campaign subsets is necessary

Consider two conflicting campaigns i and j .

	Expected per-round utility	Expected per-round cost
i	0.7	0.4
j	0.5	0.2

Per-round budget: $\rho = 0.3$.

Since i and j are in conflict, they cannot be active simultaneously.

Strategies without randomization

Deterministic choice: campaign i

To meet the budget constraint, randomize only between bidding on i (with probability p) and not bidding:

$$0.4 \cdot p + 0 \cdot (1 - p) \leq 0.3 \quad \implies \quad p \leq 0.75.$$

Hence

$$\bar{f}(\text{bidding only on } i) = 0.7 \cdot 0.75 = \boxed{0.525}.$$

Deterministic choice: campaign j

Campaign j can always be played:

$$\bar{f}(\text{bidding only on } j) = \boxed{0.5}.$$

Randomization over campaign subsets improves the optimum

Now randomize between the two feasible campaign subsets $\{i\}$ and $\{j\}$: choose

$\{i\}$ with probability p , $\{j\}$ with probability $1 - p$.

The expected per-round cost of the randomization is

$$0.4 \cdot p + 0.2 \cdot (1 - p).$$

Using the full budget:

$$0.4 \cdot p + 0.2 \cdot (1 - p) = 0.3 \implies p = 0.5.$$

Randomization over campaign subsets improves the optimum

Therefore, the expected per-round utility from randomizing over subsets $\{i\}$ and $\{j\}$ is

$$R = 0.7 \cdot 0.5 + 0.5 \cdot 0.5 = 0.35 + 0.25 = \boxed{0.6 > 0.525}.$$

Conclusion

There exists at least one setting in which randomization over feasible subsets is strictly better than any deterministic choice, even when randomization is already allowed within a campaign

⇒ We cannot choose a single deterministic subset of campaigns on which to bid in every round: our algorithm must keep a distribution over the subsets!

What can we decompose?

As we have seen, linearity lets us decompose $\bar{f}(\gamma)$ and $\bar{c}(\gamma)$, ignoring correlations in the competing bids, for example:

$$\bar{f}(\gamma) = \sum_{i=1}^N \mathbb{E}_{\mathbf{b} \sim \gamma} \left[\mathbb{E}_{m^{(i)} \sim \mathcal{D}_{\mathbf{m}_i}} \left[f^{(i)}(b_i, m_i) \right] \right]$$

Can we also ignore correlations in the joint distribution between our bids, i.e. can we simply learn one independent distribution for each campaign, sample these and then form a joint action using the sampled actions?

What can we decompose?

⇒ No! Campaigns are coupled by the conflict graph, and an independent decomposition cannot model all possible joint policies. Consider two conflicting campaigns i, j and their single-bid distributions $\gamma^{(i)}, \gamma^{(j)}$:

- 1 if we allow both their $\gamma^{(i)}$ and $\gamma^{(j)}$ to bid a non-zero value, we end up violating the conflict
- 2 if we constrain all the probability mass of either $\gamma^{(i)}$ or $\gamma^{(j)}$ to the 0.0 bid, we permanently lose the possibility of bidding over that campaign

The right abstraction: factorizing the joint probability

What can we do then?

Let $S \subseteq \{1, \dots, N\}$ be a subset of campaigns on which we bid in a round.

- Separate the superarm $\mathbf{b}$ into:
 - ① which campaigns are active;
 - ② which bid is used for each active campaign.
- A feasible superarm can then be represented as

$$a = (S, \mathbf{b}_S),$$

where S is an conflict-free subset of campaigns and $\mathbf{b}_S = (b_i \in \mathcal{B} \sim \gamma_{i|S} \text{ for every } i \in S)$, with $\gamma_{i|S}$ being the distribution over the bids of campaign i when the subset S is played.

- The expected utility and cost for any subset S are additive:

$$\bar{f}(S) = \sum_{i \in S} \bar{f}^{(i)}(\gamma_{i|S}), \quad \bar{c}(S) = \sum_{i \in S} \bar{c}^{(i)}(\gamma_{i|S}).$$

The right abstraction: factorizing the joint probability

- We can do for $\mathbf{b}$ what we did earlier for $\mathbf{m}$ in the expression for $\bar{f}(\gamma)$ (same for $\bar{c}(\gamma)$):

$$\begin{aligned}\bar{f}(S) &= \sum_{i \in S} \mathbb{E}_{\mathbf{b} \sim \gamma} \left[\mathbb{E}_{m^{(i)} \sim \mathcal{D}_{\mathbf{m}_i}} \left[f^{(i)}(b_i, m_i) \right] \right] \\ &= \sum_{i \in S} \mathbb{E}_{b_i \sim \gamma_{i|S}} \left[\mathbb{E}_{m^{(i)} \sim \mathcal{D}_{\mathbf{m}_i}} \left[f^{(i)}(b_i, m_i) \right] \right]\end{aligned}$$

This means that, even though we cannot ignore overall correlations in the distribution over the full superarms due to the conflict graph, we can at least ignore the correlations in the distribution over bids once we have fixed a feasible subset of campaigns!

$\implies$ the full joint distribution $\gamma = \mathbb{P}[\mathbf{b}]$ over the competing bids is not needed to evaluate the expected utility/cost of a superarm!

The right abstraction: factorizing the joint probability

- But: **the distribution over feasible superarms is still necessary** to encode the conflict graph and randomize across campaign subsets.

$\implies$ "post-select" between campaigns by factorizing the joint distribution γ , i.e.

$\mathbb{P}[\mathbf{b}] = \mathbb{P}[S, \mathbf{b}_S]$. Using the definition of conditional probability and the fact that we can ignore intra-subset correlations among bids of different campaigns, we have

$$\mathbb{P}[S, \mathbf{b}_S] = \mathbb{P}[S] \mathbb{P}[\mathbf{b}_S|S] = \mathbb{P}[S] \prod_{i \in S} \mathbb{P}[b_i|S]$$

We can thus define a two-step sampling procedure for the joint action:

- 1 sample a conflict-free subset S of campaigns from a distribution $\gamma_S = \mathbb{P}[S]$ over all the feasible subsets;
- 2 sample one bid for each campaign from the conditional distributions $\gamma_{i|S} = \mathbb{P}[b_i|S]$.

Clairvoyant formulation: factorized distribution

$$\begin{aligned}
 \gamma_s^*, [\gamma_i]^* &= \arg \max_{\gamma_s \in \mathbb{R}^{2^N}, [\gamma_{i|S}] \in \mathbb{R}^{NK}} \sum_S \sum_{i \in S} \sum_b \gamma_s(S) \gamma_{i|S}(b) \bar{f}^{(i)}(b) \\
 \text{s.t.} \quad & \sum_S \sum_{i \in S} \sum_b \gamma_s(S) \gamma_{i|S}(b) \bar{c}^{(i)}(b) \leq \rho && \text{(budget)} \\
 & \sum_S \gamma_s(S) = 1 && \text{(subset probability)} \\
 & \sum_b \gamma_{i|S}(b) = 1 \quad \forall S, i && \text{(conditional probability)} \\
 & \gamma_{i|S}(0) = 0 \quad \forall S, \forall i \in S && \text{(campaign in subset)} \\
 & \gamma_{i|S}(0) = 1 \quad \forall S, \forall i \notin S && \text{(campaign not in subset)} \\
 & \gamma_s(S) = 0 \quad \forall S \text{ having conflicts} && \text{(conflict)} \\
 & 0 \leq \gamma_s(S) \leq 1 \quad \forall S \\
 & 0 \leq \gamma_{i|S}(b) \leq 1 \quad \forall b
 \end{aligned}$$

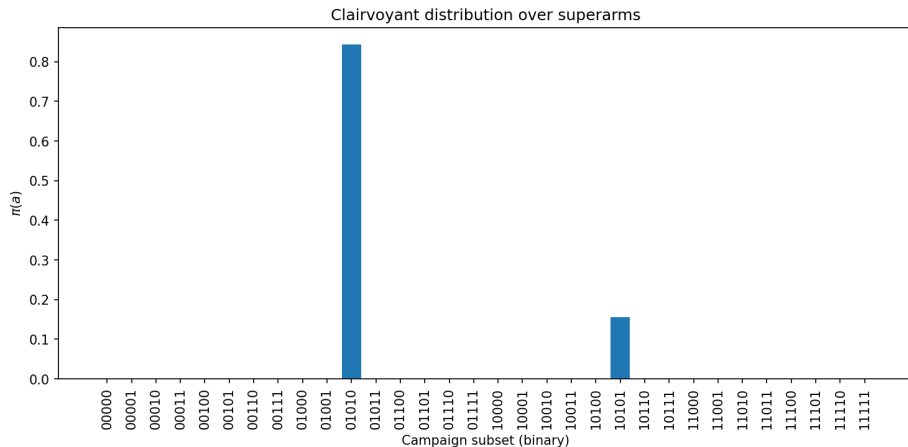
$\Rightarrow$ Non-linear Program!

Clairvoyant formulation: factorized distribution

Redefine the LP variables into $\gamma_{S,i}(b)$ to use the joint probability $\mathbb{P}[S, b_i] = \gamma_S(S)\gamma_{i|S}(b)$ instead (same amount of variables) $\Rightarrow$ **Linear Program!**

$$\begin{aligned}
 \gamma_S^*, [\gamma_i]^* &= \arg \max_{\gamma_S \in \mathbb{R}^{2^N}, [\gamma_{S,i}] \in \mathbb{R}^{NK}} \sum_S \sum_{i \in S} \sum_b \gamma_{S,i}(b) \bar{f}^{(i)}(b) \\
 \text{s.t.} \quad & \sum_S \sum_{i \in S} \sum_b \gamma_{S,i}(b) \bar{c}^{(i)}(b) \leq \rho && \text{(budget)} \\
 & \sum_S \gamma_S(S) = 1 && \text{(subset probability)} \\
 & \sum_b \gamma_{S,i}(b) = \gamma_S(S) \quad \forall S, i && \text{(marginalization)} \\
 & \gamma_{S,i}(0) = 0 \quad \forall S, \forall i \in S && \text{(campaign in subset)} \\
 & \gamma_{S,i}(0) = 1 \quad \forall S, \forall i \notin S && \text{(campaign not in subset)} \\
 & \gamma_S(S) = 0 \quad \forall S \text{ having conflicts} && \text{(conflict)} \\
 & 0 \leq \gamma_S(S) \leq 1 \quad \forall S \\
 & 0 \leq \gamma_{S,i}(b) \leq 1 \quad \forall b
 \end{aligned}$$

Clairvoyant: what γ^* over subsets of campaigns looks like



A randomized distribution over all the subsets of campaigns optimized by the clairvoyant of Requirement 2. We do not show the conditional distributions over bids within the single campaigns, since they can differ from one subset to another.

Combinatorial-UCB: algorithm

Implements the UCB-like bidding approach for a combinatorial set of joint actions (i.e., bids on multiple campaigns)

Core idea

UCB estimates + conflict-aware campaigns subset optimization + budget constraint

- **Exploration rounds (first K rounds):** randomly-chosen feasible superarms, aiming to explore as many arms as possible
 - Some arms might not be chosen $\implies$ no problem, confidence radius gets set to 1 so that they will be explored afterwards
- **Action selection:** we could, as in the single-campaign setting, solve the same LP as the clairvoyant and then sample a bid from the resulting optimal distribution...

Combinatorial-UCB: simpler approach

...but it's kinda expensive to do that *every single round*! Instead, we split the clairvoyant LP into two suboptimal (but simpler) Linear Programs:

- First, we solve the single-campaign LP from Requirement 1 with per-round budget ρ to obtain an optimal distribution $\gamma^{(i)}$ over bids for each campaign;
- Then, we use the solutions $\gamma^{(i)}$ to form expected utilities/costs for each campaign, and we select a distribution γ_s^* over campaigns subsets through the following LP over all the campaign subsets:

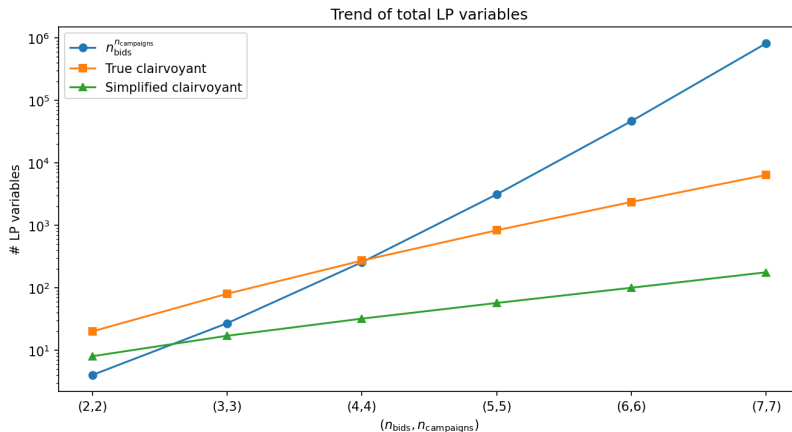
$$\begin{aligned}
 \gamma_s^* = \arg \max_{\gamma_s \in \mathbb{R}^{2^N}} \quad & \sum_S \gamma_s(S) \left(\sum_{i \in S} \bar{f}^{(i)}(\gamma^{(i)}) \right) \\
 \text{s.t.} \quad & \sum_S \gamma_s(S) \left(\sum_{i \in S} \bar{c}^{(i)}(\gamma^{(i)}) \right) \leq \rho && \text{(budget)} \\
 & \sum_S \gamma_s(S) = 1 && \text{(subset probability)} \\
 & \gamma_s(S) = 0 \quad \forall S \text{ having conflicts} && \text{(conflict)} \\
 & 0 \leq \gamma_s(S) \leq 1 \quad \forall S
 \end{aligned}$$

True clairvoyant against simplified clairvoyant

We can also use this approach to build a clairvoyant; however, experimental results show that this approach achieves a little less expected utility than the true clairvoyant that we explained earlier $\implies$ for this reason, we use this approach only in the Combinatorial-UCB algorithm, but we keep the true clairvoyant for computing the regret

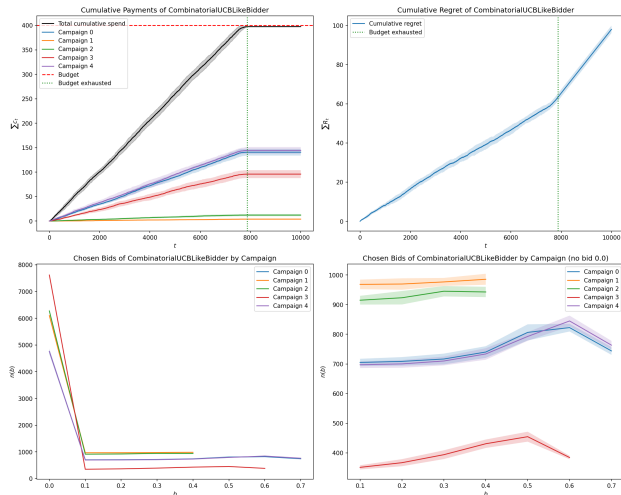
```
True clairvoyant expected utility: 0.016285210741043048
Simplified clairvoyant expected utility: 0.015386339054128496
True clairvoyant expected payment: 0.039999999999999994
Simplified clairvoyant expected payment: 0.04
Increase in true clairvoyant utility: 0.000898871686914552
Increase in true clairvoyant payment: -6.938893903907228e-18
```

Explosion of LP variables



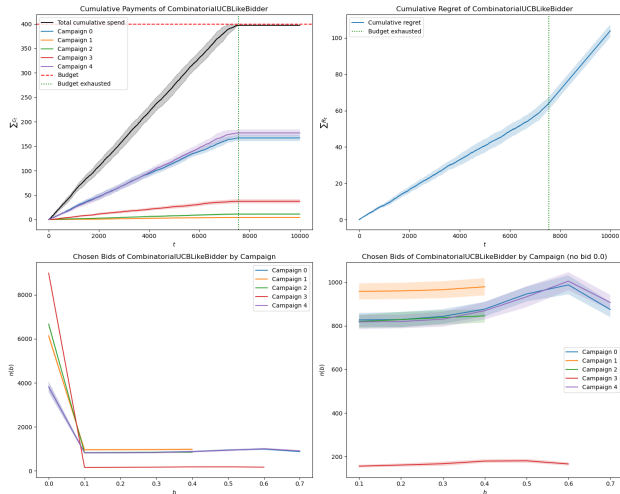
Comparison of the variables in the three possibilities for the LP: **full joint distribution** (LP not analyzed in these slides), **distribution from true clairvoyant**, **distribution from simplified clairvoyant**. Note that the y-axis is logarithmic.

Results: Combinatorial-UCB using true LP



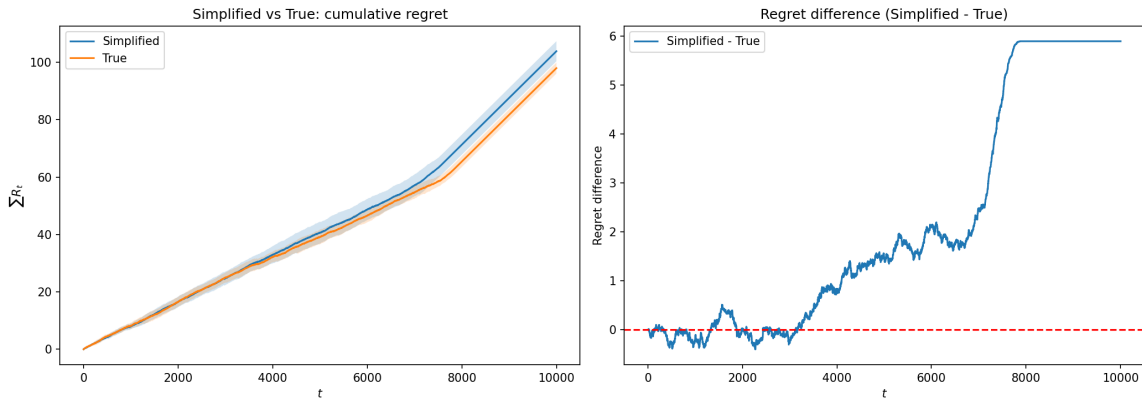
- 10000 (!) rounds are not enough to get evident sublinear regret and a clear separation among bids, although a hint of a peak among the bids of some campaigns can be seen
- Randomizing among subsets of campaigns leads to each campaign not being played sometimes

Results: Combinatorial-UCB using simplified LP



- The regret is similar to that achieved using the true LP
- The separation between bids played is even less clear

Results: comparison of the regrets of the two variants



The Combinatorial-UCB bidder that uses the true LP achieves a little less regret than the bidder using the simplified LP (but also takes three times as much time to execute $\Rightarrow$ probably not worth it!).

Requirement 3: highly non-stationary multiple-campaigns environment

Highly non-stationary multiple-campaigns environment

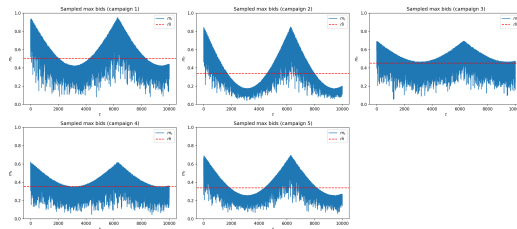
- Sequence of highest competing bids $(m_t^{(i)})_t$ sampled from a distribution that *changes quickly and continuously over time*
- Different patterns per campaign but possibly *correlated* through market regimes (e.g., holidays, Black Friday)
- UCB-style methods break: confidence radius assumes stationarity
- Algorithm to be studied: **primal-dual** with budget, assuming full feedback
 - Full-feedback: highest competing bid $m_t^{(i)}$ is observed, providing information about all arms regardless of which one we bid on
 - $\implies$ a regret minimizer for the adversarial expert problem (**Hedge**) is appropriate!

Distribution model: adversarial-style sinusoids

Modeling choice:

- Each competitor's bid at round t is drawn from $\text{Unif}(0, u_t^{(i)})$ where

$$u_t^{(i)} = \text{low}^{(i)} + (\text{high}^{(i)} - \text{low}^{(i)})(1 - |\sin(5t/T)|)$$



- Same frequency across campaigns: models price changes due to the *shared market conditions*, like sales (Black Friday), holidays (Christmas), ... during which showing ads is more valuable and competitors are likely to bid more
- high* and *low* represent the range of the oscillations
- Max of uniforms is Beta-distributed *within each round*, but the parameters change continuously, so the Beta distribution is actually rescaled

Clairvoyant

- We want to use a similar LP to that of the clairvoyant for Requirement 2
- The previous clairvoyants used the true win probabilities $\mathbb{P}[m \leq b]$ from the known underlying stationary distribution
- But: now, the distribution of the highest competing bids keeps changing quickly throughout the rounds $\implies$ no fixed win probabilities!

Instead, use the estimates $\hat{\mathbb{P}}[m \leq b]$ of the empirical win probabilities (computed in hindsight)!

Assumption

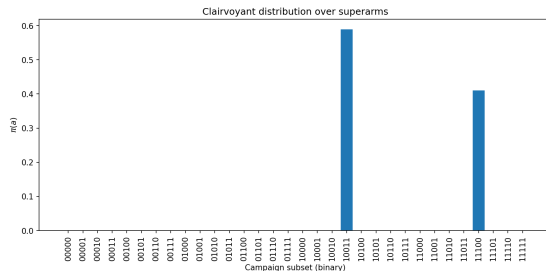
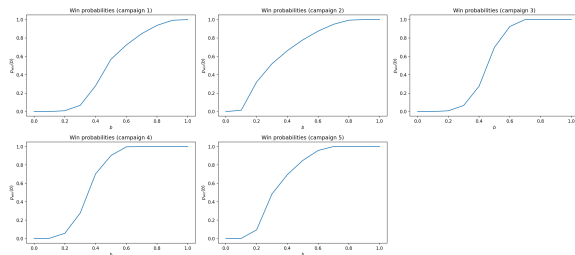
The clairvoyant knows the empirical win probability $\hat{\mathbb{P}}[m \leq b]$ a posteriori of each bid b (like in the exercise session).

Clairvoyant: from Requirement 2 to Requirement 3

$$\begin{aligned}
 \gamma_s^*, [\gamma_i]^* &= \arg \max_{\gamma_s \in \mathbb{R}^{2^N}, [\gamma_{S,i}] \in \mathbb{R}^{NK}} \sum_S \sum_{i \in S} \sum_b \gamma_{S,i}(b) \bar{f}^{(i)}(b) \\
 \text{s.t.} \quad & \sum_S \sum_{i \in S} \sum_b \gamma_{S,i}(b) \bar{c}^{(i)}(b) \leq \rho && \text{(budget)} \\
 & \sum_S \gamma_s(S) = 1 && \text{(subset probability)} \\
 & \sum_b \gamma_{S,i}(b) = \gamma_s(S) \quad \forall S, i && \text{(marginalization)} \\
 & \gamma_{S,i}(0) = 0 \quad \forall S, \forall i \in S && \text{(campaign in subset)} \\
 & \gamma_{S,i}(0) = 1 \quad \forall S, \forall i \notin S && \text{(campaign not in subset)} \\
 & \gamma_s(S) = 0 \quad \forall S \text{ having conflicts} && \text{(conflict)} \\
 & 0 \leq \gamma_s(S) \leq 1 \quad \forall S \\
 & 0 \leq \gamma_{S,i}(b) \leq 1 \quad \forall b
 \end{aligned}$$

The win probabilities affect the LP from within $\bar{f}^{(i)}(b) = (v - b) \mathbb{P}_{m \sim \mathcal{D}_m} [b \geq m]$ (and $\bar{c}^{(i)}(b)$): simply replace $\mathbb{P}_{m \sim \mathcal{D}_m} [b \geq m]$ with $\hat{\mathbb{P}}_{m \sim \mathcal{D}_m} [b \geq m]$!

Clairvoyant: optimal γ under highly non-stationarity



Left: empirical $\hat{\mathbb{P}}[b \geq m_t]$, estimated a posteriori over all rounds.

Right: optimal strategy γ^* of the clairvoyant, which still randomizes over multiple campaigns subsets.

Primal-dual bidder: optimization problem

Remember:

γ is the distribution over joint actions, induced in our case by the two-step sampling procedure

The optimization problem that we are solving is

$$\gamma^* = \arg \max_{\gamma \in \Delta_{\mathcal{B}}} \bar{f}(\gamma) \quad \text{s.t. } \bar{c}(\gamma) \leq \rho$$

The probability simplex constraint is already enforced by the choice of Hedge as regret minimizer: the budget constraint is instead incorporated through the expected Lagrangian

$$\bar{\mathcal{L}}(\gamma, \lambda) = \underbrace{\bar{f}(\gamma)}_{\text{expected utility}} - \lambda \left(\underbrace{\bar{c}(\gamma)}_{\text{expected payment}} - \rho \right)$$

where $\bar{f}(\gamma) = \sum_{i=1}^N \mathbb{E}_{\mathbf{b} \sim \gamma} \left[\mathbb{E}_{(m^{(1)}, \dots, m^{(N)}) \sim \mathcal{D}_{\mathbf{m}}} [f^{(i)}(b_i, m_i)] \right]$ (and same for $\bar{c}(\gamma)$)

Primal-dual bidder: structure

- **Dual step (stochastic gradient ascent on λ):**

$$\lambda_{t+1} = \Pi_{[0,1/\rho]} (\lambda_t - \eta_d (\rho - c_t))$$

in which c_t is estimating $\bar{c}(\gamma)$ in the expected Lagrangian

Primal-dual bidder: structure

- **Primal step (multiplicative Hedge updates on γ):** $N + 1$ separate Hedge instances on
 - N per-campaign bid distributions γ_i over $\mathcal{B} \setminus \{0.0\} \rightarrow$ Hedge updates:

$$w_{t+1}^{(i)}(b) = w_t^{(i)}(b) e^{-\eta_p \ell_t^{(i)}(b)} \quad \forall b \in \mathcal{B}$$

- campaigns-subsets distribution $\gamma_s \rightarrow$ Hedge updates:

$$w_{t+1}(S) = w_t(S) e^{-\eta_p \mathcal{L}_t(S)} \quad \forall S \in \mathcal{P}(\mathcal{N})$$

Same decomposition as the Combinatorial-UCB bidder in Requirement 2!

Conflicts are masked by initializing to 0 the weights corresponding to invalid superarms:
Hedge's multiplicative updates won't be able to change these weights

Primal-dual bidder: structure

- **Primal step (multiplicative Hedge updates on γ):**

$$w_{t+1}^{(i)}(b) = w_t^{(i)}(b) e^{-\eta_p \ell_t^{(i)}(b)} \quad \forall b \in \mathcal{B}$$

$$w_{t+1}(S) = w_t(S) e^{-\eta_p \mathcal{L}_t(S)} \quad \forall S \in \mathcal{P}(\mathcal{N})$$

We actually use two kinds of Lagrangians for the primal step: one "global" Lagrangian $\mathcal{L}_t$ (like the dual one) for Hedge over the campaigns subsets, and single-campaign Lagrangians ℓ_t for the Hedge instances over the single campaigns

Primal-dual bidder: Lagrangians

At round t , we observe the highest competing bid $m_t^{(i)}$ for all bids in all campaigns:

- $f_t^{(i)}(b_i) = (v_i - b_i) \mathbb{1} [b_i \geq m_t^{(i)}]$ is the utility obtained from campaign i if playing bid b_i
- $c_t^{(i)}(b_i) = b_i \mathbb{1} [b_i \geq m_t^{(i)}]$ is the payment for campaign i if playing bid b_i

The overall expected utility and payment for any subset S of campaigns are

$$f_t(S) = \sum_{i \in S} \mathbb{E}_{\mathbf{b} \sim \gamma_{i|S}} [f_t^{(i)}(b_i)] = \sum_{i \in S} f_t^{(i)}(\gamma_{i|S}), \quad c_t(S) = \sum_{i \in S} \mathbb{E}_{\mathbf{b} \sim \gamma_{i|S}} [c_t^{(i)}(b_i)] = \sum_{i \in S} c_t^{(i)}(\gamma_{i|S}).$$

Primal-dual bidder: Lagrangians

With the budget constraint $c_t \leq \rho$, the global per-round Lagrangian is

$$\mathcal{L}_t(S, \lambda_t) = f_t(S) - \lambda_t(c_t(S) - \rho)$$

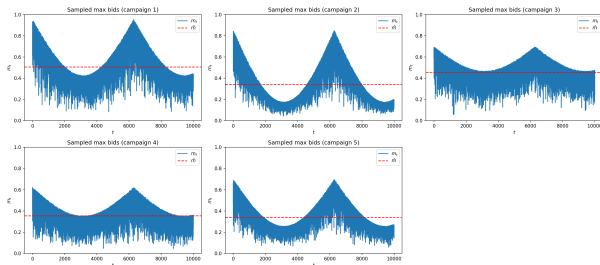
Key point

The term $\lambda_t \rho$ appears **once**, regardless of how many campaigns ($|S|$) are in subset S

The single-campaign Lagrangians, instead, only use $f_t^{(i)}(b_i)$ and $c_t^{(i)}(b_i)$ of one campaign:

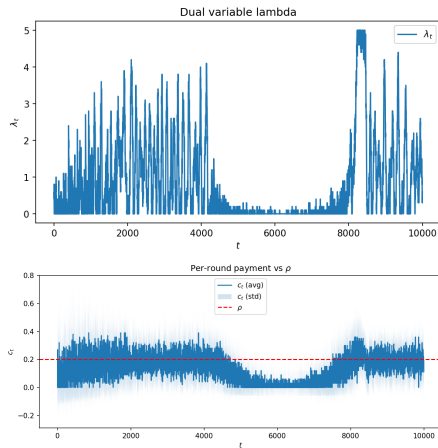
$$\ell_t^{(i)}(b_i, \lambda_t) = f_t^{(i)}(b_i) - \lambda_t(c_t^{(i)}(b_i) - \rho)$$

Results: dual variable λ_t and per-round expense

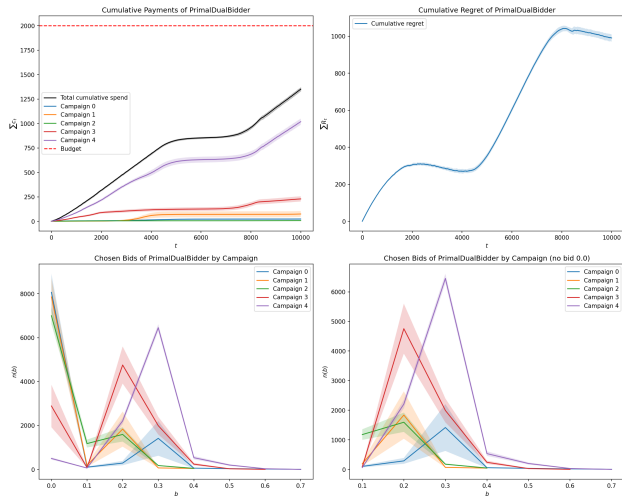


λ_t drifts upward or downward (**right: top plot**) respectively when the bidder spends less or more than ρ (**right: bottom plot**).

- During the peaks in the **left plot**, λ_t decreases, counteracting the need of spending more due to the overall higher competing bids;
- during the valleys, lambda grows, because the current auctions being less expensive means that we could try to win them more often.



Results: primal-dual



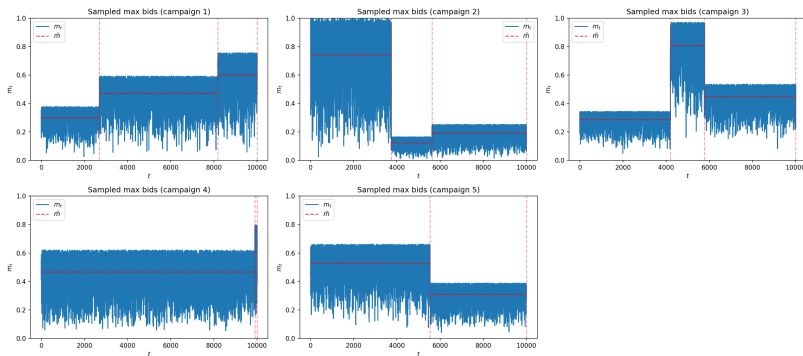
- The primal-dual bidder is very conservative and spends much less than the budget
- This happens whatever the budget is: this bidder never depletes all the budget (observed while making attempts, but not in the notebook)
- Regret is sublinear, but clearly follows the trend in the highest competing bids
- Bids are (surprisingly) clearly separated: the primal-dual bidder likely acts on the pacing of the bids (how often), not on the value of the bid itself

Requirement 4: slightly non-stationary multiple-campaigns environment

Slightly non-stationary multiple-campaigns environment

- Rounds partitioned in *stationary intervals* (phases), each with its own joint distribution of highest competing bids $m_p^{(i)}$
- Algorithms to be compared:
 - **Combinatorial-UCB with sliding window**
 - **Combinatorial-UCB with CUSUM change detection**
 - **Primal-dual** (same as Req 3, no change awareness)

Distribution model: piecewise-stationary phases



- For each campaign i , independently sample the number of phases $\Upsilon_i \sim \text{Unif}\{2, 3\}$ and then $\Upsilon_i - 1$ change times uniformly in $[1, T]$
- Each phase has a different upper bound $u_p^{(i)} \sim \text{Unif}(0, 1)$ on competitors' uniform bids
- Within a phase: competing bids $\sim \text{Unif}(0, u_p^{(i)})$, thus $m_p^{(i)} \sim \text{Beta}(n_i, 1)$ but *rescaled by* $u_p^{(i)}$
 $\implies$ closed-form win probabilities $\implies$ we *can* write the same LP of the stochastic setting, per-phase

Clairvoyant: realistic baseline

- Slightly non-stationary clairvoyant: the one implementing the *best policy* in hindsight (defined in the exercise session)
- Actual best policy in a budget-constrained environment: deliberately do not bid at all in some phases, wait for later phases in which highest competing bids are lower instead.
- $\implies$ Completely unrealistic: no learner can foresee that a more favourable phase than the current one is coming in the future!
 - (See proof of lower bound for adversarial environment in the course slides)
- Thus, we choose the clairvoyant **not as knowing all the changes in the highest competing bids in advance**, but **as being able to detect the changes instantaneously as they happen**
 - In line with our learners: they "lose rounds" in detecting changes, or in trying to forget the past, or in adjusting against them

Clairvoyant: a per-phase LP on each interval

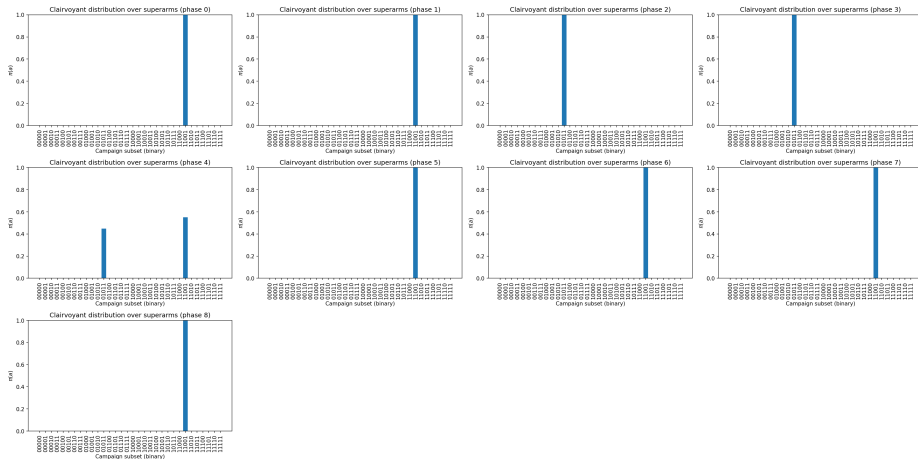
Intersect the phases of all campaigns together and split rounds in a single *joint* partition

A clairvoyant simply needs to solve a stationary LP for each phase in this joint partitioning!
 $\implies$ For each joint phase p : solve the same clairvoyant LP as in Requirement 2, with phase-specific win probabilities $\mathbb{P}_p^{(i)}(b \geq m)$

Note:

ρ stays the same across phases, it does not get updated!

Clairvoyant: phase-wise γ_s^* over campaigns subsets



Each phase has its own randomization policy over the subsets of campaigns: the policy changes because some campaigns become better/worse in different phases due to the competing bids being non-stationary.

Sliding-Window Combinatorial-UCB: recap

- Maintain a *window* of the samples played over the last τ rounds and the corresponding utilities/costs observed
- Confidence radius is computed on the window only: $r(b) = \sqrt{\frac{2 \log \tau}{N_\tau(b)}}$
- Theoretically optimal window size: $\tau = \sqrt{\frac{T \log T}{\Upsilon}}$, where Υ is the total amount of (*joint!*) phases
 - for unknown Υ , set $\Upsilon = T \implies \tau = \sqrt{\log T}$
 - $\tau \approx \sqrt{T}$ is also a good choice in practice: this is what we use
 - we actually use $\tau \approx \sqrt{T \log T}$
- Same underlying bidder as Combinatorial-UCB from Requirement 2

CUSUM Combinatorial-UCB: recap

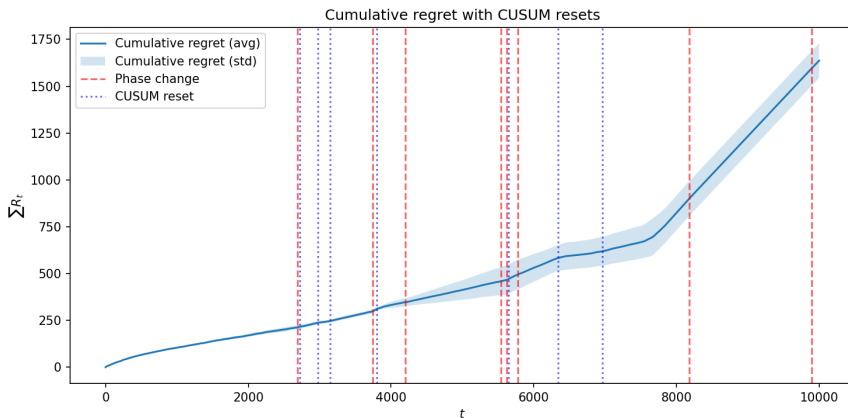
Two phases:

- Estimation phase: build an estimator for the average utility/cost of each arm
 - For each campaign, we set an exploration counter to a value M , and we keep exploring arms until all the counters reach 0
 - We try to select superarms that explore arms that require estimation for multiple campaigns at the same time
 - At the same time, we do not bid on arms that we have already estimated to avoid wasting budget, since they may no longer be optimal
- Detection phase: detect significant deviations from the estimated mean, which signal a phase change
 - We compute the positive/negative deviation of the current utility/cost from the estimated means, "dampening" by a quantity ε
 - We keep track of the positive/negative cumulative deviations from the estimates across multiple rounds
 - We detect a change if either is larger than a threshold h

CUSUM Combinatorial-UCB: recap

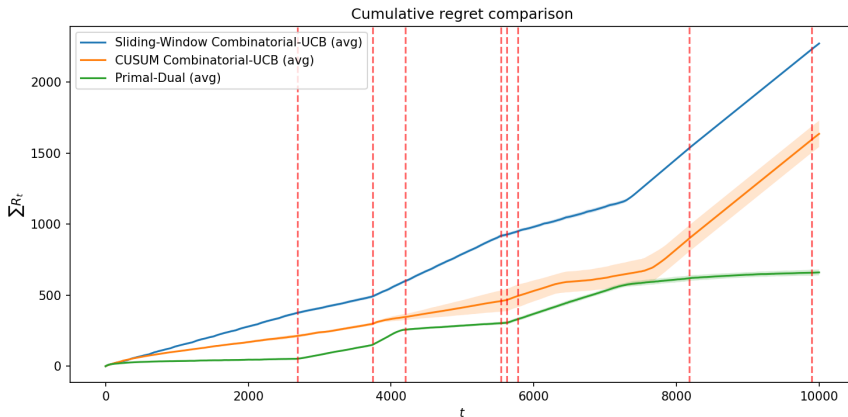
- Extra probability α of pure exploration per-round to detect changes in arms not being played
- Hyperparameters to be tuned:
 - M : Length of the estimation phase
 - ε : "Dampening" factor, how much we expect a sample to deviate from the mean
 - h : Threshold for detecting a change
 - α : Probability of "pure" exploration per-round
- Again, same underlying bidder as Combinatorial-UCB from Requirement 2

CUSUM in action: detection traces



Joint phase change times (vertical red lines) and CUSUM reset times (vertical blue lines) superimposed on the cumulative-regret curve. Regret can be convex/nonsmooth when changing phase, like the phase change right before round 4000, or the ones right before round 6000. A CUSUM reset time can lead to concavity and improvements in the regret, like in the two resets after round 6000.

Comparison: cumulative regret of the bidders



Unexpectedly, the primal-dual bidder remains the best, with CUSUM Combinatorial-UCB being a surprisingly close second, until it depletes the budget. Sliding-Window Combinatorial-UCB does a bit worse, maybe due to a suboptimal choice of the window size. Still, all regrets exhibit sublinear behavior until budget is depleted.

Recap of the findings

One-slide recap

Setting	Best fit	Regret
Stochastic, single campaign	UCB1-like	Sublinear
Stochastic, multi campaign	Combinatorial-UCB	Sublinear*
Highly non-stationary	Primal-dual	Sublinear
Slightly non-stationary	CUSUM-UCB	Sublinear

*For Combinatorial-UCB, many more rounds are required, or confidence intervals decaying faster

- **Clairvoyant** baseline needs to be adapted carefully to the problem, must be neither too strong (unrealistic) nor too weak (non-informative)
- **Hyperparameters** should be tuned carefully, especially for CUSUM and Sliding-Window methods
- **Expensiveness** of the bidder is non-negligible, especially when rounds progress with a fast pace

Thanks!

Questions?

Code, notebooks and results:

<https://github.com/Richypiants/online-learning-applications-project-2025-2026/>